Manuscript

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Contents

Disentangling contaminants from true intratumoural microbial signals in pancreatic ductal adenocarcinoma – A benchmarking approach **1**
Abstract . . . 1
Introduction . . . 2
Results . . . 2
Discussion . . . 14
Materials and Methods . . . 15
References . . . 15

Disentangling contaminants from true intratumoural microbial signals in pancreatic ductal adenocarcinoma – A benchmarking approach

Abstract

Background: Malignant tumors or more specifically Pancreatic ductal adenocarcinoma (PDAC) harbors their own distinct intratumoral microbiome, yet rigorous characterization of its composition is hampered by pervasive environmental and procedural contamination. Negative controls collected throughout the sample processing workflow capture a heterogeneous mixture of environmental, and technician-derived microbes that, if unaccounted for, mislead 16S rRNA sequencing results.

Methods: We systematically collected more than 200 negative control samples (NCT) comprising buffer, paraffin, and PCR controls, over period of four years and other possible significant batches such as years, seasons, technicians. We applied and benchmarked four decontamination methods (Restrictive filtering, Decontam, SCRuB, and the Nejman pipeline) against this NCT-derived contaminant profile, using both a composite score. Further, we validated those methods via technical replicas with Aitchison distance between them as orthogonal evaluation metrics.

Results: Microbial profiles of NCT samples were significantly determined by control type, technician, year, and season, reflecting complex batch effects. The 20 most abundant contaminant taxa spanned well-characterized environmental genera (e.g., *Sphingomonas*) and human commensals (e.g., *Veillonella parvula*). Decontamination benchmarking demonstrated that the Nj method consistently outperformed alternatives in both composite score and inter-replicate concordance. Application of Nj to fresh frozen PDAC samples substantially reduced contaminant burden while preserving putative tumor-associated signals.

Conclusions: This benchmarking framework, combining longitudinal NCT profiling with technical replication, enables principled assessment of decontamination performance in the absence of a ground truth. Our results support the adoption of the Nj decontamination pipeline for future intratumoral microbiome studies in PDAC and provide a reliable, quality control strategy for low-biomass 16S sequencing studies.

Keywords: pancreatic ductal adenocarcinoma, tumor microbiome, decontamination, negative controls, 16S rRNA sequencing, benchmarking.

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Note: significant and new literatures should be mentioned:

- Salzberg 2026 (1) - Setting higher standards for reports of microbial species in human cancers
- Cite (2) SCRuB paper for benchmarking on simulation data, as well as in clinical data, but highlight its limitation.
- Highlight our Nj procedure adaptation, using Fisher test instead of Binomial test, due to small number of samples. (3)

Introduction

The intratumoral microbiome has emerged as a potentially important component of the tumor microenvironment, influencing immune evasion, drug metabolism, and clinical outcomes across multiple cancer types [citations]. In pancreatic ductal adenocarcinoma (PDAC), one of the most lethal malignancies, recent studies have reported the presence of distinct intracellular bacterial communities that may contribute to disease pathogenesis and immunosuppression [citations]. However, a fundamental challenge in tumor microbiome research is distinguishing genuine tumor resident microorganisms from environmental contaminants introduced during sample collection, processing, and sequencing. Low biomass specimens such as FFPE tissue sections and tumor biopsies are particularly vulnerable to contamination, as trace microbial DNA from reagents, laboratory surfaces, and personnel can overwhelm endogenous taxa. Prominent environmental taxa such as **Sphingomonas, Ralstonia, and Pseudomonas** are routinely detected in negative controls and have been reported as dominant species in tumor microbiome studies that lack rigorous decontamination (4, 5).

Several computational decontamination approaches have been proposed to address this problem, including Decontam (6), SCRuB (2), and the strategy employed by Nejman et al. (7) (hereafter, Nj). These methods differ in their statistical assumptions, required inputs, and aggressiveness of taxon removal. Nevertheless, a systematic, empirical comparison of their performance in the context of PDAC tumor microbiome profiling has been lacking. Here, we present a benchmarking framework that leverages a longitudinal collection of negative control samples (NCT) acquired across all major wet lab processing steps, combined with technical replicates of PDAC tissue samples. We evaluate four decontamination strategies using a novel composite score, balancing taxon yield and purity relative to the NCT-derived contaminant profile, and validate findings using inter-replicate Aitchison distances. This work provides practical guidance for microbiome researchers seeking to generate reliable intratumoral microbial profiles from low-biomass clinical specimens.

Results

Structure for result section

- Contaminant species survey in the lab
  - Description of data collection, number of samples and taxa, significant batch effects.
  - Alpha/beta analyses.
  - Barplot composition / heatmap plot of bacterial composition of NCT.
  - DAA among control types
    - Tables and figures
    - Description
  - Remarks
- Hood vs. bench environments
  - Experimetal setting
  - Alpha/beta analyses, batch effect
  - DAA w.r.t technicians
- Decontamination method benchmarking
  - Using bacterial profile before/after decontamination and contaminants survey for assessment
  - Distance between technical replicas for assessment

- Intratumoral microbiome in PDAC with replicas
  - Each decontamination method, microbial composition corresponding.
  - Composition of overlap between different methods
  - Composition overlap between replicas for each method.
- (FFPE restroration)
- Suggested pipeline
  - For 16S sequencing with ONT
  - Fresh forzen tissue instead of FFPE
  - Involve negative controls, technical replica, Bioinformatics: batch correction and decontamination

1. Contamination Survey in the Laboratory

1.1 Negative Control Samples Collection To comprehensively characterize the contaminant landscape, we collected negative control (NCT) samples at each stage of the wet-lab processing pipeline (Fig. 1.1), including 93 paraffin controls, 133 buffer controls, 11 PCR controls, and 3 sequencing controls, yielding an initial pool of 6,128 taxa. Sequencing controls were excluded due to insufficient read depth (750 reads). Taxa with extremely low prevalence or abundance were removed (See Methods section), and samples were normalized via Wrench or rarefication. After filtering, 203 NCT samples remained: 113 buffer controls, 84 paraffin controls, and 6 PCR controls, encompassing 1,775 taxa after rarefaction and 2,305 taxa prior to rarefaction. These controls were accumulated between 2021 and 2024, with a sharp expansion of paraffin and buffer control types introduced in 2022 (Table 1). Relevant batch covariates include processing technician, year, and season.

1.2 Alpha and Beta Diversity of Negative Controls Significant differences in microbial diversity were observed across control types (Figs. 1.2, 1.3). Buffer controls exhibited significantly higher species richness and evenness compared to paraffin and PCR controls, as measured by observed species count, Shannon index, and inverse Simpson index. Analogous patterns were observed when stratifying by technician, year, and season (Supplementary Figures **S1.xxx, S1.xxx**), suggesting pervasive batch structure in the contaminant pool.

1.3 Microbial Composition of Negative Control Samples The 20 most abundant taxa identified across all NCT samples consisted of a mixture of well characterized environmental microbes and small portion of human commensals (Figs. 1.4, 1.5). For example, *Sphingomonas*, a ubiquitous environmental genus frequently detected in hospital and laboratory environments, was among the most abundant taxa in nearly all negative controls. In contrast, the human commensal *Veillonella parvula*, previously reported in PDAC-associated microbiome studies (8), was present in approximately 60% of NCT samples but at considerably lower abundance (Fig. 1.4). This coexistence of environmental and human commensal taxa in negative controls requires a need for nuanced decontamination strategies; indiscriminate removal of all NCT-associated taxa risks eliminating clinically relevant tumor microbiome signals.

Following the correction for batch effects, differential abundance analysis (DAA) among control types revealed several taxa specifically enriched in paraffin controls (taxa **XXX** from the analyses, Figure 1.6), pointing to contamination sources specific to paraffin embedding procedures.

Table 1. Negative control samples collected across years, stratified by control types.

Sample ↓ / Feature →	2021	2022	2023	2024
Buffer	29	60	24	0
Paraffin	5	57	11	12
PCR	0	3	3	0

Figure 1.1. Schematic of negative control sample collection at each stage of the wetlab processing pipeline (top). Summary of potential significant batch effects associated with negative control samples (bottom).

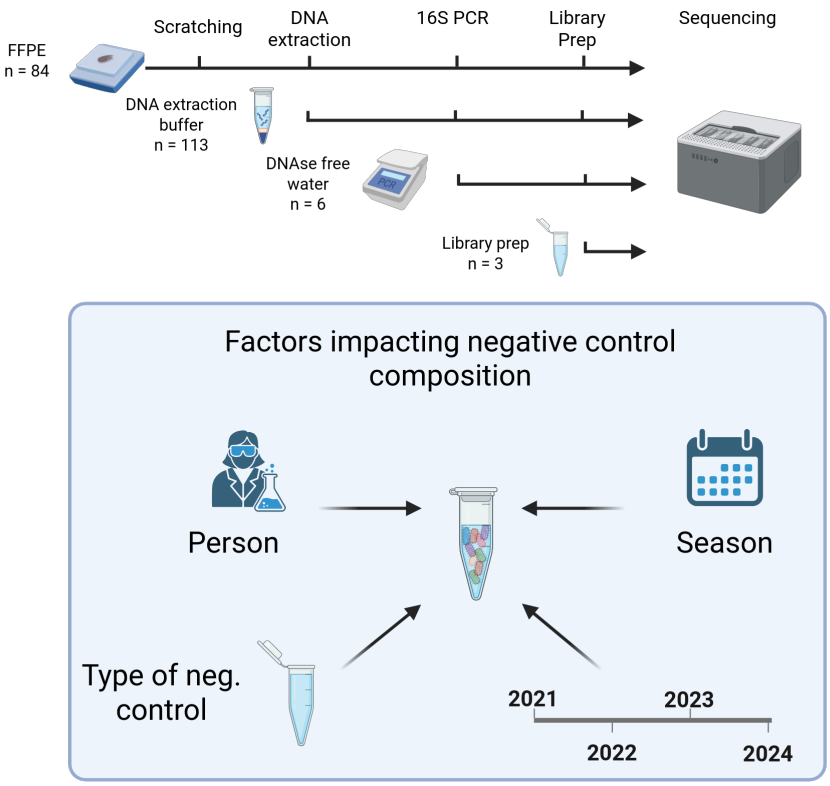
FFPE
n = 84
Scratching
DNA
extraction
16S PCR
Library
Prep
Sequencing
DNA extraction
buffer
n = 113
DNAse free
water
n = 6
Library prep
n = 3
Factors impacting negative control
composition
Person
Season
Type of neg.
control
2021
2022
2023
2024

Figure 1:

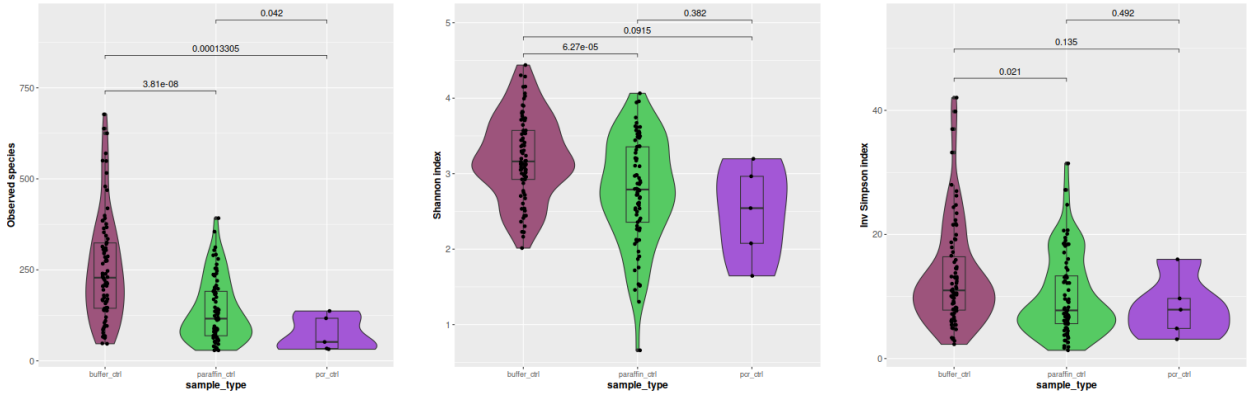

Figure 1.2 A–C) Alpha diversity of NCT samples by control type: observed species richness (A), Shannon index (B), and inverse Simpson index (C). Statistically significant pairwise differences are indicated.

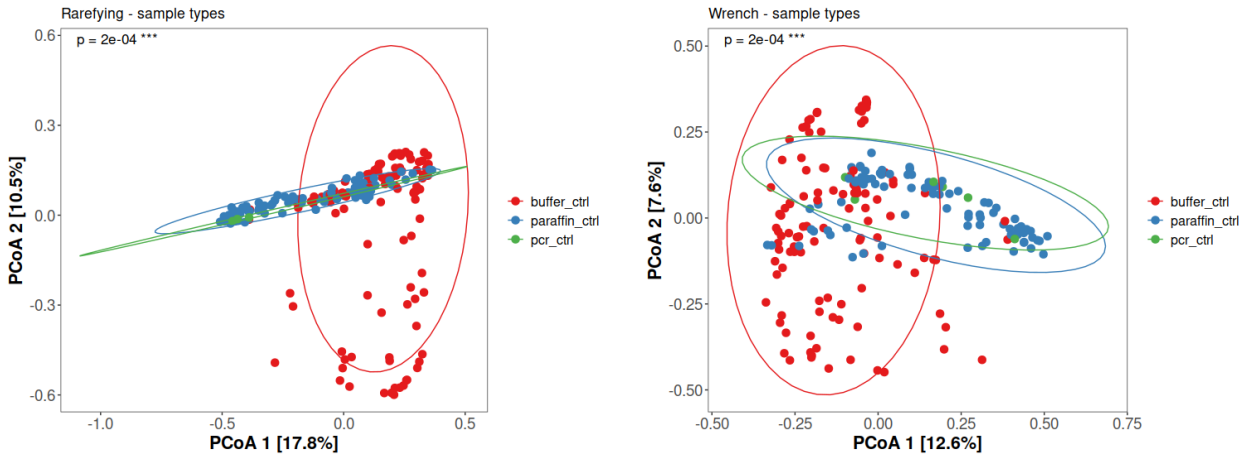

Figure 1.3 (A–B) Beta diversity ordination of NCT samples using rarefaction (A) and Wrench normalization (B). Group centroids and PERMANOVA p-values are shown.

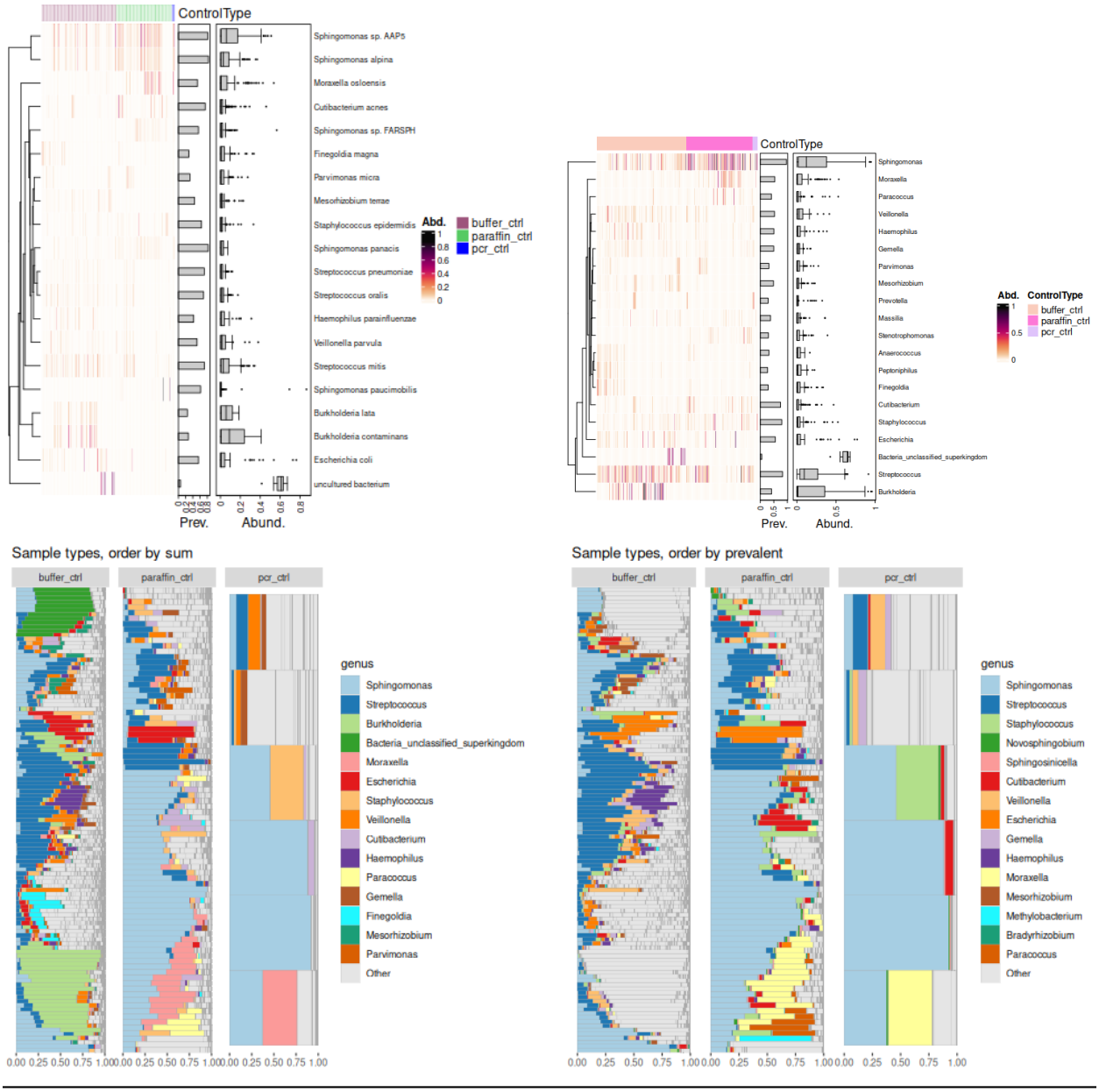

Figure 1.4 Heatmap of the top 20 most abundant taxa in NCT samples at species (left) and genus (right) level. Bottom panels show barplots of the top 20 genera ordered by abundance and prevalence, respectively.

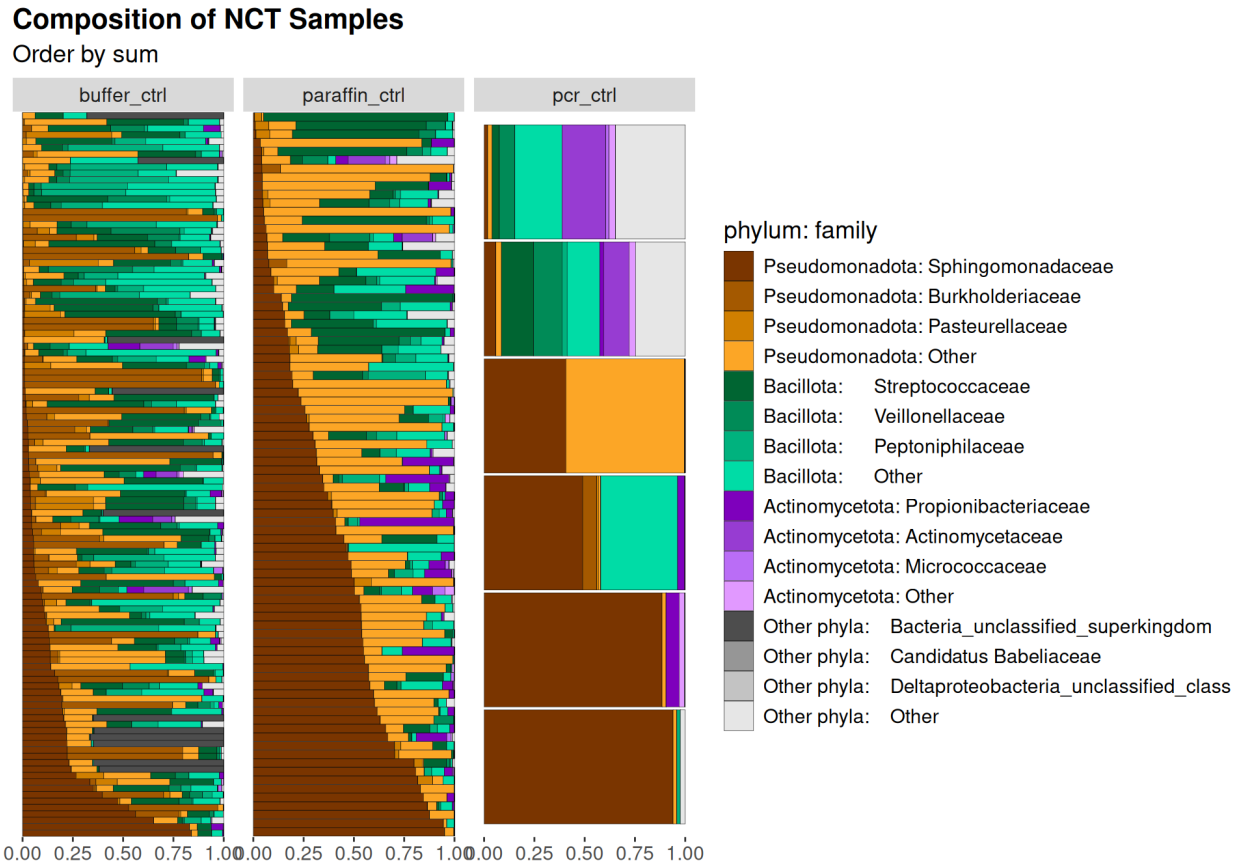

Figure 1.5 Stacked barplot of bacterial composition in negative control samples, stratified by control type.

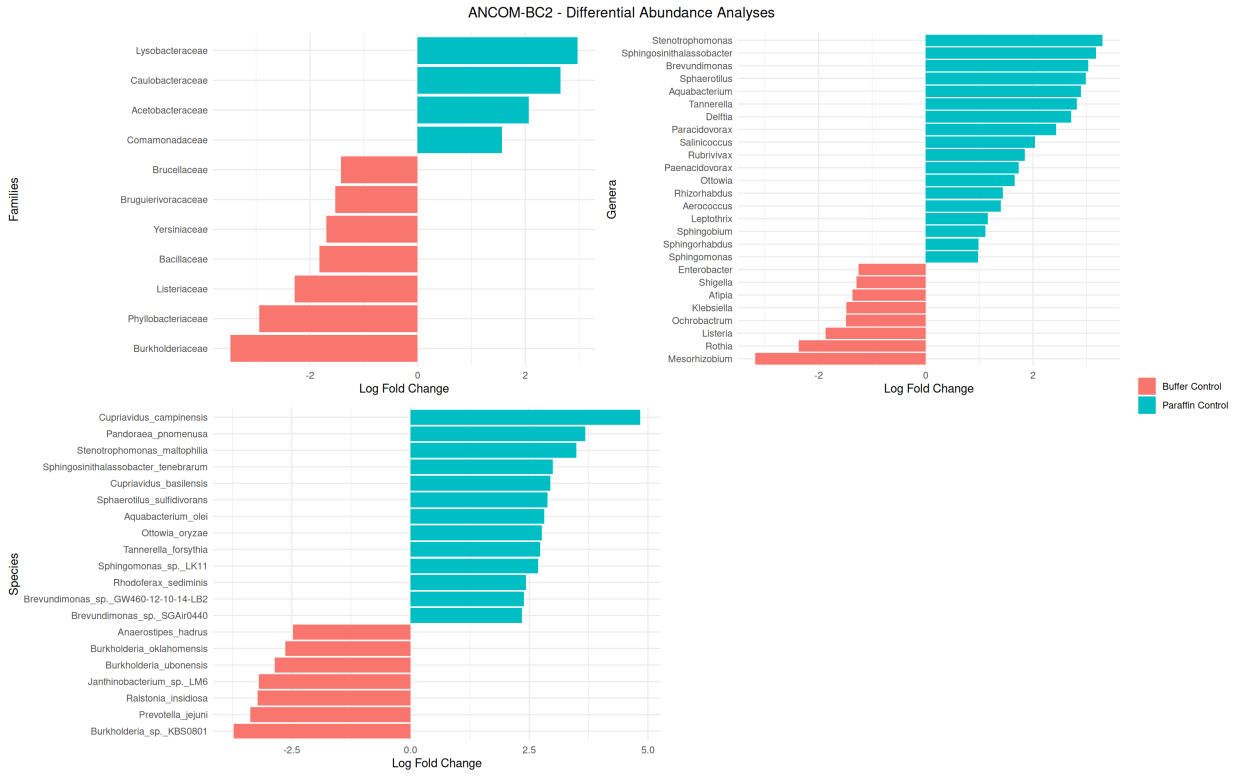

Figure 1.6 Differential abundance analyses via ANCOM-BC2 across various taxonomic rank between paraffin and buffer control types after acconting for technician batch correction.

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2. Technician Influences Microbiome More Than Processing Environment

To investigate the origin of human commensal taxa in NCT samples, we designed a controlled experiment in which two technicians processed negative control samples across two environmental conditions: a standard bench environment and a laminar flow hood (Fig. 2.1; see Materials and Methods for full protocol). Alpha diversity analysis revealed no significant difference with respect to environmental condition; however, a clear distinction between technicians was observed (Fig. 2.2). Beta diversity analysis similarly demonstrated significant clustering by technician identity (PERMANOVA test - p-value < 0.05), with no significant effect of environmental conditions (p-value $= 0.6$; Fig. 2.3). Notably, after adjusting for technician, the bacterial profiles of clean (hood) and standard (bench) conditions diverged significantly ($p = 0.02$), indicating that while technician-derived contamination is dominant, environmental setting contributes an independent, detectable signal.

Given the low read counts and high sparsity of NCT data, DAA between conditions was performed using ANCOM-BC2 (9) and ALDEx2 (10) with relaxed significance thresholds (ANCOM-BC2: $q < 0.1$; ALDEx2: $p < 0.2$). These analyses identified a set of taxa consistently enriched under bench conditions, consistent with human skin and oral commensals introduced during opening bench handling (Fig. 2.4).

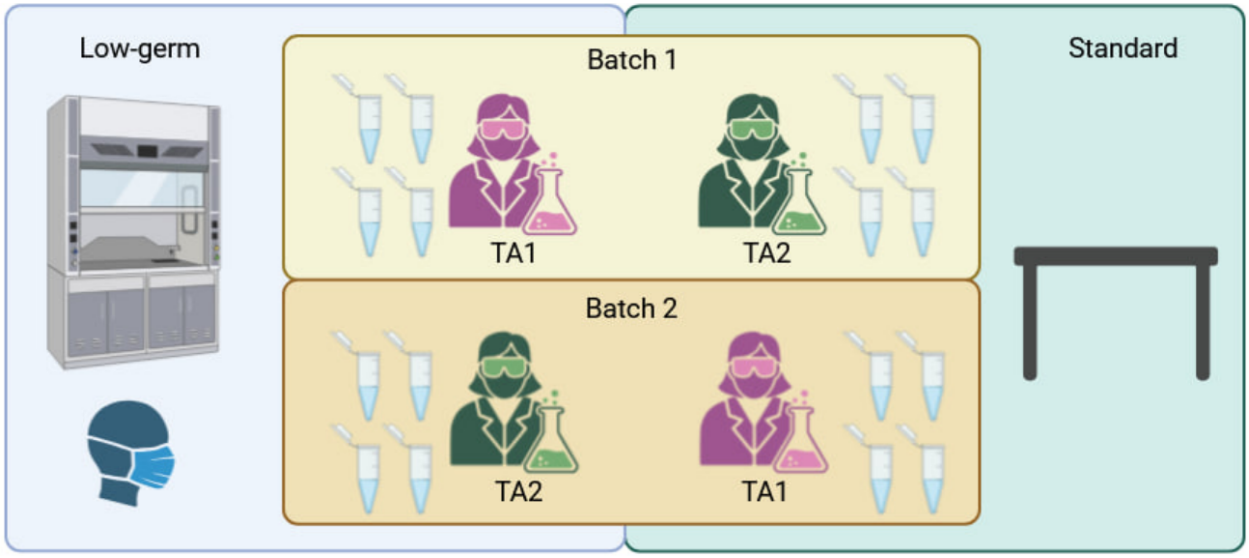

Figure 2.1 Schema of the experimental setting where two technicians processed negative control samples across different environments.

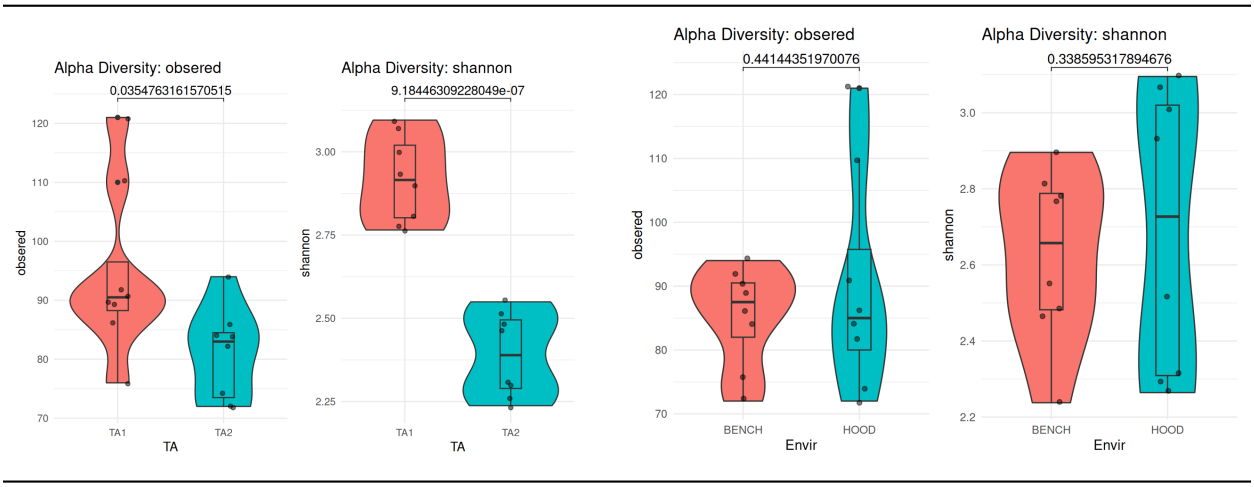

Figure 2.2 Alpha diversity of NCT samples stratified by technician (left) and environmental conditions(right). We used number of observed species as well as Shannon index as metrics for alpha analyses.

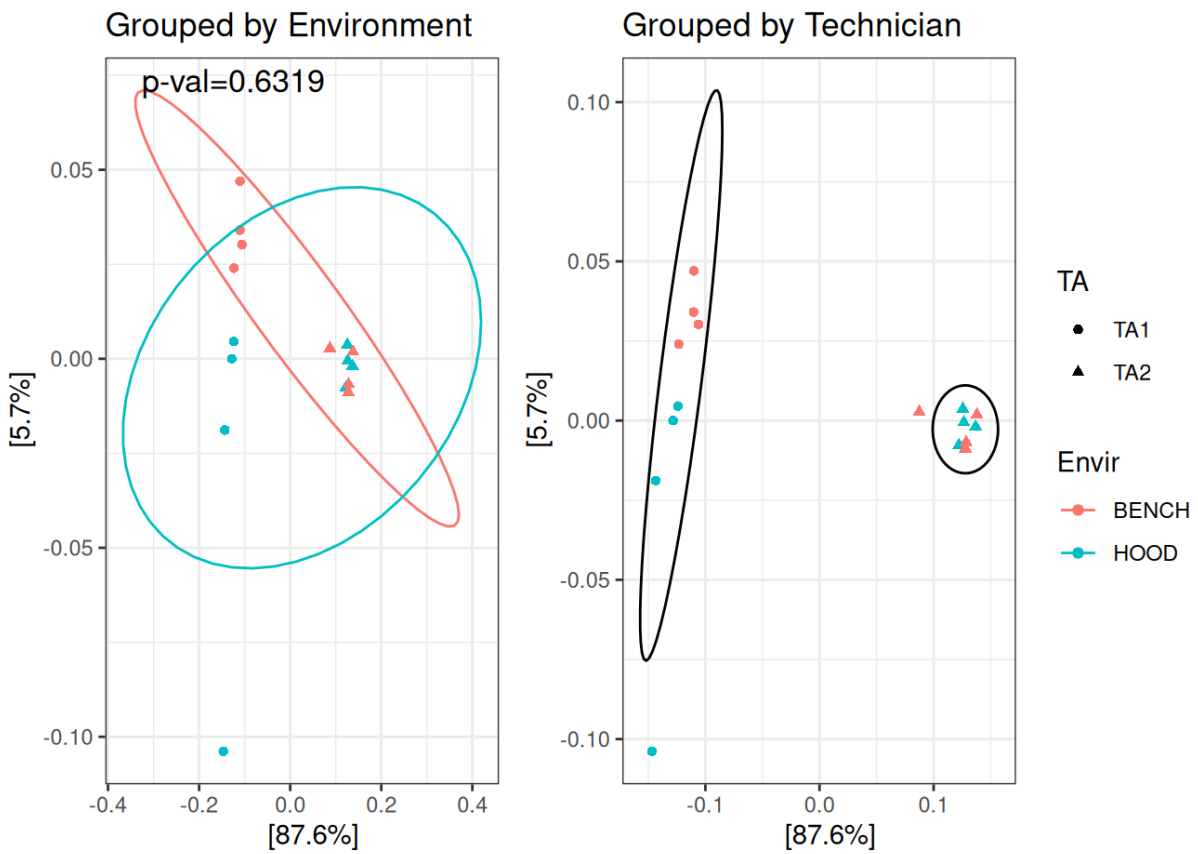

Figure 2.3 Beta diversity cluster by technicians (left) and environmental conditions (right). Colors depict different environmental conditions, while shapes depict different technicians.

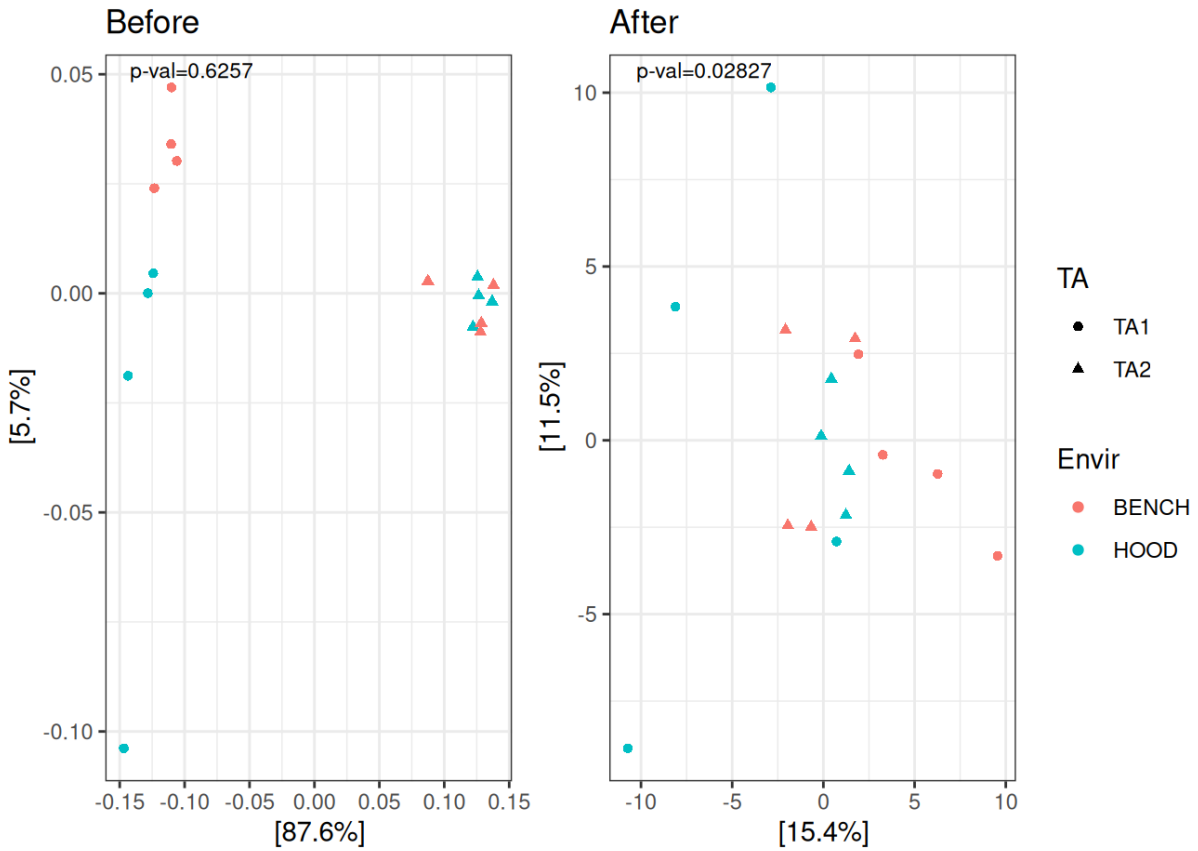

Figure 2.3+ (perhaps for supplemtary)

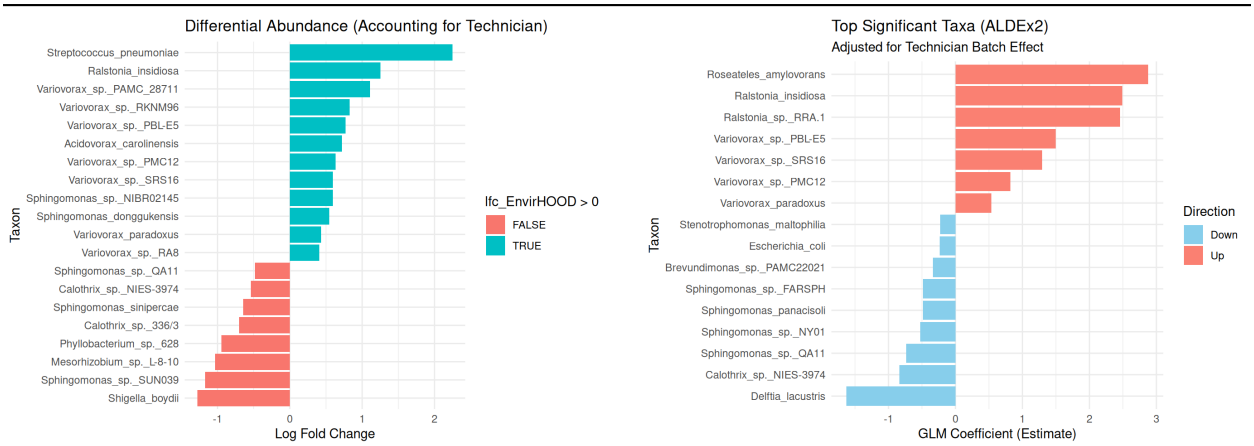

Figure 2.4 Differential abundance analyses between normal (bench) and clean (hood) environments by ANCOM-BC (left) and ALDEx2 (right) respectively.

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3. Benchmarking of Decontamination Methods

3.1 Assessment Using Longitudinal Negative Control Survey We applied four decontamination strategies, including Restrictive filtering, Decontam R package (6), SCRuB (2), and Nj procedure (7) to

the 16S profiles of eighteen fresh frozen PDAC samples, generating four decontaminated abundance tables alongside the unprocessed raw data (retained as a baseline). The benchmarking of various decontamination strategies conducted through simulation and clinical dataset has been reported elsewhere (2). Nevertheless, simulation data could not reflect the complexity of extreme low biomass of intratumoral samples under overwhelmed contaminated environments. Moreover, the human-derived samples in (2) do not encounter the extreme low biomass cubersome such as intratumoral PDAC samples we had, thus not sutable for our assessment. In the absence of a ground truth, we leveraged the longitudinally derived contaminant profile (Section 1) as a negative reference for evaluation. We defined a composite score comprising two complementary components: yield and purity. First, yield is the proportion of putative significant taxa absent from the NCT survey over the number of observed species. On the other hand, purity is the proportion of NCT-absent taxa among all putative significant taxa over the total number of putative significant taxa. The composite score, which is the product of yield and purity, simultaneously rewards methods that exclude contaminant taxa while prevents excessive removal of true taxa (see Materials and Methods section for more details). Scores were computed per sample, and methods were compared using a linear mixed model to account for sample level random effects. The Nj method achieved the highest composite score, significantly outperforming all alternatives (Figure 3.1). Untreated raw data scored lower than Nj and restrictive method, on the same level with Decontam method. Interestingly, the composite score from SCRuB is the smallest. Normally SCRuB shows its effectiveness from the leakage well information, which we do not have in our experimental settings.

3.2 Assessment Using Technical Replicates To provide an orthogonal validation, we re-sequenced ten PDAC fresh frozen samples with the highest read counts as technical replicates. Our underlying assumption was that decontaminated profiles, freed from random contaminant variation, should exhibit greater concordance between replicates from the same tumor than raw profiles.

Inter-replicate dissimilarity was quantified using Aitchison distance, with pairwise comparisons assessed by paired Wilcoxon signed-rank test. As shown in Figure 3.2, the Nj method again obtained the smallest inter-replicate distances, indicating superior within sample consistency. On the other hand, inter-replicate among samples without decontamination show the highest scores, indicating that environmental contaminants could be heterogeneous. Other decontamination methods yielded smaller distances than raw data, but still significantly bigger than Nj method.

Furthermore, compositional overlap between replicas, measured as the fraction of shared taxa out of total detected taxa between two replicates was also highest for Nj (Table 2). This measurement estimates the robustness of decontamination procedure. Note that we excluded the extremly rare taxa and aggrogate data into genus rank level.

Table 2. Overlap: fraction of shared taxa over total taxa among replicates, counted in genus rank, with min_prevalence = 5% and min_total_abundance = $1e-4$ from tax_filter function of microViz R package (11).

Methods ↓ / Features →	n_rep1	n_rep2	n_intersect	n_union	fraction
Raw	140	106	71	175	0.406
Restrictive	116	95	66	145	0.455
Decontam	124	102	70	156	0.449
SCRuB	136	105	74	167	0.443
Nj	63	65	50	78	0.641

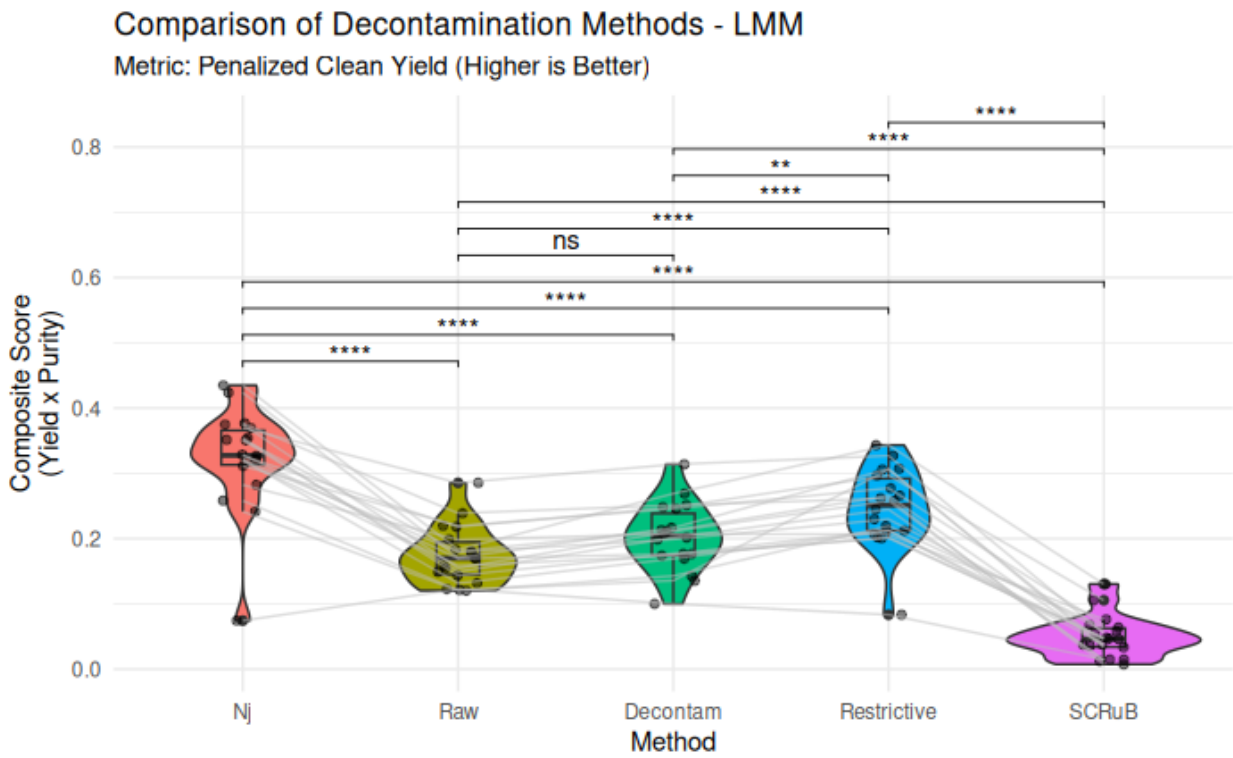

Figure 3.1 Composite score was calculated for each sample, statistical testing via Linear Mixed Model

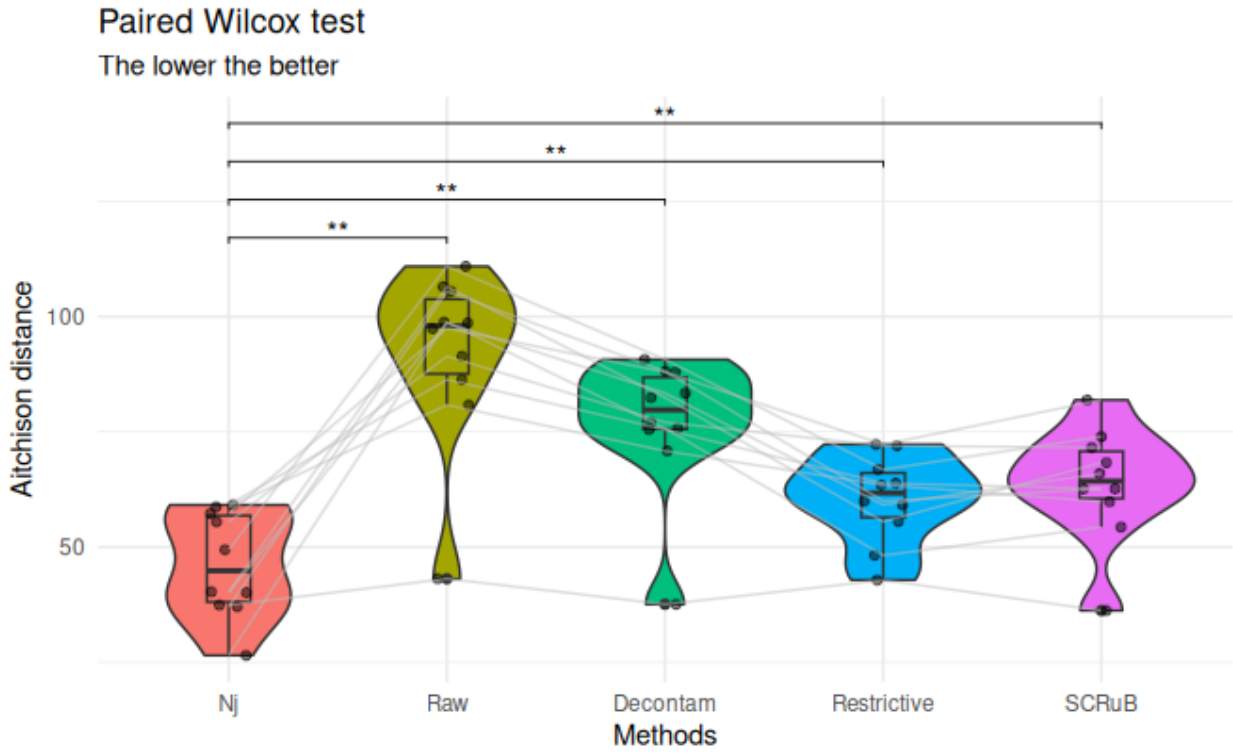

Figure 3.2 Atchiton distance between replicates, statistical test with paired Wilcox ranked test.

4. Intratumoral Bacterial Profile of PDAC Fresh Frozen Samples

We applied all decontamination strategies to characterize the intratumoral bacterial community of the ten PDAC fresh-frozen samples. In the raw, unprocessed data, samples were dominated by environmental contaminants. For example, Sphingomonas predominated in first replicates of several samples, while Caldibacillus was prominent in second replicates of others (Fig. **XXX**). These taxa were largely eliminated following decontamination.

Despite rigorous decontamination, the bacterial profiles of technical replicates from the same tumor were not identical, consistent with the known spatial heterogeneity of intratumoral microbiome distributions (12). Nevertheless, decontamination could be able significantly reduce inter-replicate Aitchison distances relative to raw data (Fig. 3.2), supporting the conclusion that residual compositional differences reflect biological variation rather than technical noise. Following Nj decontamination, several taxa with prior biological relevance to PDAC were consistently detected across replicates, including [**taxa to be specified upon analysis completion**].

Discussion

In this study, we present a systematic benchmarking framework for assessing decontamination methods in low biomass 16S sequencing of PDAC tumor microbiome. By combining a longitudinal negative control survey with technical replicate sequencing, we establish two complementary: data driven evaluation metrics, a composite yield-purity score and Aitchison inter-replicate distance which do not require a priori knowledge of true contaminants.

Our negative sample survey revealed that the laboratory contaminant landscape is shaped by multiple interacting batch factors, including control types, processing technicians, year, and season. Crucially, human commensals such as Veillonella parvula were present in a majority of NCT samples, challenging the common practice of using any NCT-detected taxon as a definitive contaminant label. Instead, thorough and multi-level filtering such as (7) shows roburst ability to selectively remove likely contaminants while preserving taxon diversity, a balance that is critical when studying the sparse, low abundance microbial communities characteristic of pancreatic tumor tissue.

The Nj decontamination method consistently outperformed Restrictive filtering, Decontam, and SCRuB across all evaluation metrics. This advantage likely stems from Nj's incorporation of both prevalence-based and rigorous statistical test, as well as its conservative approach to taxon removal. Our findings align with the original validation of this strategy in Nejman et al. (2020), while extending it to a standardized benchmarking context with independent replication.

Several limitations should be considered. First, the NCT profiles used as reference contaminants may not capture all sources of contamination present in intratumoral samples, particularly those introduced at tissue collection. Second, the small sample size (n = 18, and n=10 with replicates) limits statistical power for detecting rare intratumoral taxa and constrains generalizability. Third, FFPE samples, which comprise the majority of archival PDAC biobanks, were not included in the study due to restoration complexity and overwhelmed by environmental contaminants, which beyond the capacity of intratumoral microbiome restoration.

Based on our findings, we recommend the following quality control pipeline for 16S sequencing regarding the intratumoral microbiome studies in PDAC: (1) include negative controls representing all wet-lab processing steps; (2) acquire technical replicates, at least for a subset of samples; (3) consider batch correction along with decontamination; and (4) apply various decontamination method, followed by validation of residual taxa using the composite yield-purity framework and iter-replicates distances described here. Fresh frozen samples should be prioritized over FFPE where feasible, as FFPE restoration introduces additional sources of DNA degradation and chemical modification that complicate microbial profiling.

Materials and Methods

Sample Collection and Processing

16S rRNA Sequencing

Normalization

Trivial Filtering Taxa by Prevalence and Abundance

Statistical Analysis

Decontamination Methods

Four decontamination approaches were evaluated: (1) Restrictive filtering: taxa detected in any NCT samples were excluded from further analyses; (2) Decontam (6): probability-based classification of contaminant taxa using negative control prevalence; (3) SCRuB (2): source tracking-based contaminant removal; and (4) Nejman et. al. procedure (7): the decontamination strategy incorporating multiple filtering criteria.

Composite Score

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